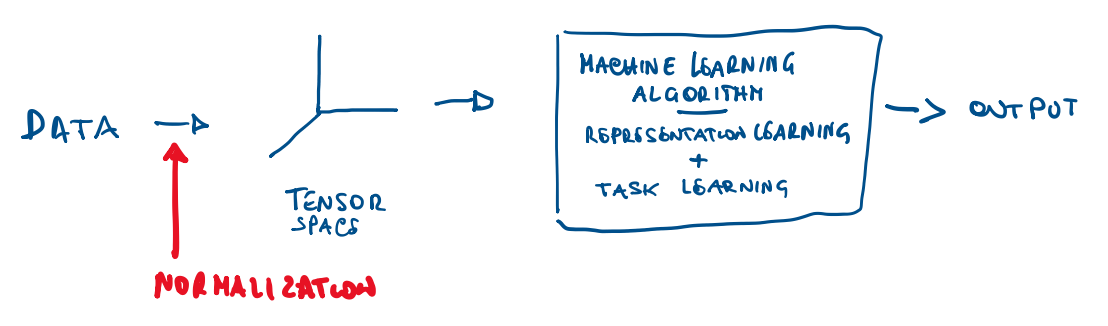
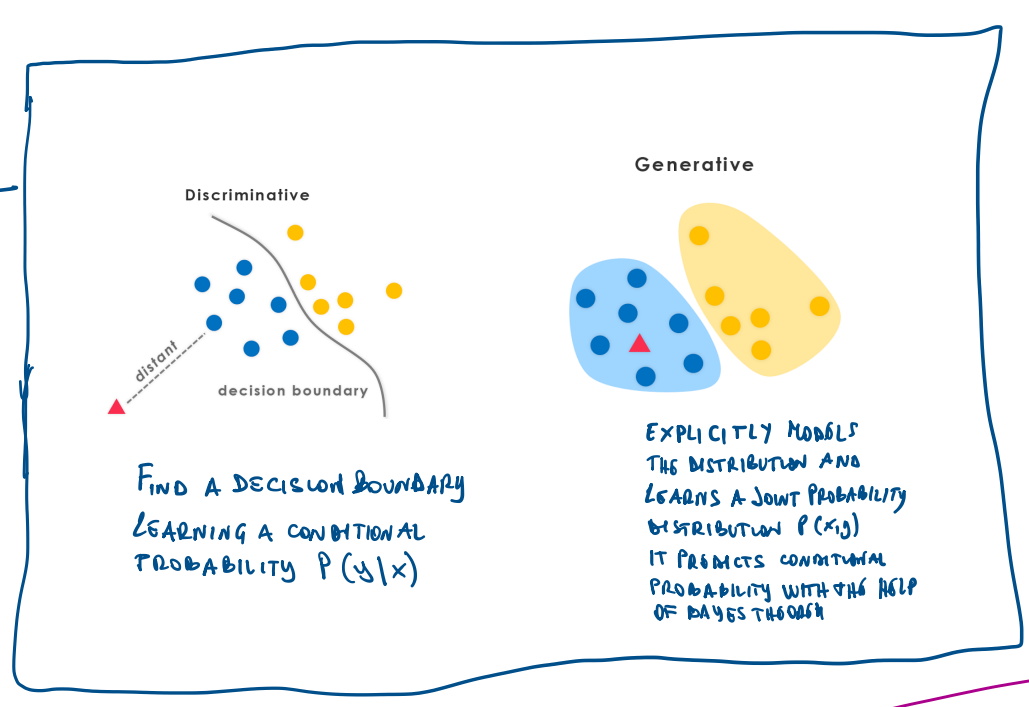
# PIPELINE IN ML


DATA
TENSOR SPACE
MACHINE LEARNING ALGORITHM
REPRESENTATION LEARNING
+
TASK LEARNING
OUTPUT
NORMALIZATION

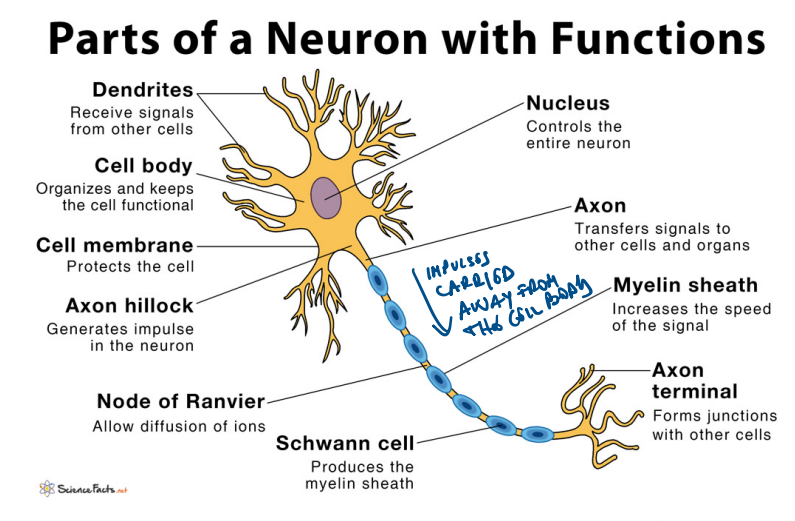
## GENERATIVE VS. DISCRIMINATIVE MODELS


Discriminative
distant
decision boundary
Generative

FIND A DECISION BOUNDARY LEARNING A CONDITIONAL PROBABILITY $P(y|x)$

EXPLICITLY MODELS THE DISTRIBUTION AND LEARNS A JOINT PROBABILITY DISTRIBUTION $P(x,y)$
IT PREDICTS CONDITIONAL PROBABILITY WITH THE HELP OF BAYES THEOREM

## ORIGIN OF NEURAL NETWORKS AND DEEP LEARNING


Parts of a Neuron with Functions

Dendrites
Receive signals from other cells
Cell body
Organizes and keeps the cell functional
Cell membrane
Protects the cell
Axon hillock
Generates impulse in the neuron
Node of Ranvier
Allow diffusion of ions
Schwann cell
Produces the myelin sheath
Nucleus
Controls the entire neuron
Axon
Transfers signals to other cells and organs
IMPULSES CARRIED AWAY FROM THE CELL BODY
Myelin sheath
Increases the speed of the signal
Axon terminal
Forms junctions with other cells

SUM

**1958**
PERCEPTRON
(ROSENBLATT)
RESAMBLE A NEURON
- ADJUST PARAMETERS WHITH FIXED ± 1
- METHOD TO LEARN: ADJUST PARAMETERS

**1960**
ADELINE
(WIDROW AND HOFF)
ADAPTIVE LINEAR NEURON
PHISICAL DEVICE
ADJUST PARAMETERS WITH WEIGHTED SUM OF THE INPUT

**1969**
PERCEPTRONS
(MINSKY AND PAPERT)
MATEMATICAL PROOF OF CONVERGENCE AND LIMITATIONS
(EG. XOR FUNCTION + CONNECTEDNESS → MLP)

AI WINTER

**1970-1976**
BACKPROPAGATION
(LINNAINMAA 1970)
(WERBOS 1974)
(RUMELHART, HINTON, WILLIAMS 1986)
COMPUTATIONAL METHOD USEFUL TO LEARN MLP

**1997**
LSTM
(HOCHREITER, SCHMIDHUBER)
TEMPORAL DATA, LANGUAGE SEQUENCES

**1998**
CNN (LENET-5)
(LECUN
BOTTOU
BENGIO
HAFFNER)
FOR OCR

Convolutional Network Demo from 1993 (LENET-1 1993)

**2006**
DEEP LEARNING
(HINTON, OSINDERO, TEH)
- NO GPUs AT THAT TIME
- NO BIG DATASET FOR TRAINING
- OVERFITTING
- VANISHING GRADIENT PROBLEMS (WITH NN YOU NEED TO MULTIPLY SEVERAL SMALL NUMBERS... DO NOT HELP LEARNING PROCESS)
  - BACKPROPAGATION + LAYER BY LAYER TRAINING

**2009**
IMAGE NET
(DENG, FEI-FEI)
FIRST BIG SET OF DATA LABELLED. 16M IMAGES ANNOTED BY HUMANS. 1000 DIFFERENT CLASSES

**2013**
ALEXNET
(KRIZHEVSKY, SUTSKEVER, HINTON)
RE-IMPLEMENTATION OF LENET-5 (1998)
TRAINING ON IMAGENET
THEORETICAL IMPROVEMENTS
- ACTIVATION FUNCTION (RELU)
- DROPOUT
- DATA AUGMENTATION
GOOD RESULTS IN MANY DOMAIN (IMAGES, SPEECH...)

DEEP LEARNING GOLDEN ERA

**2015**
MSRA
RESNET (154 LAYERS)

**2016**
DEEPMIND
ALPHAGO

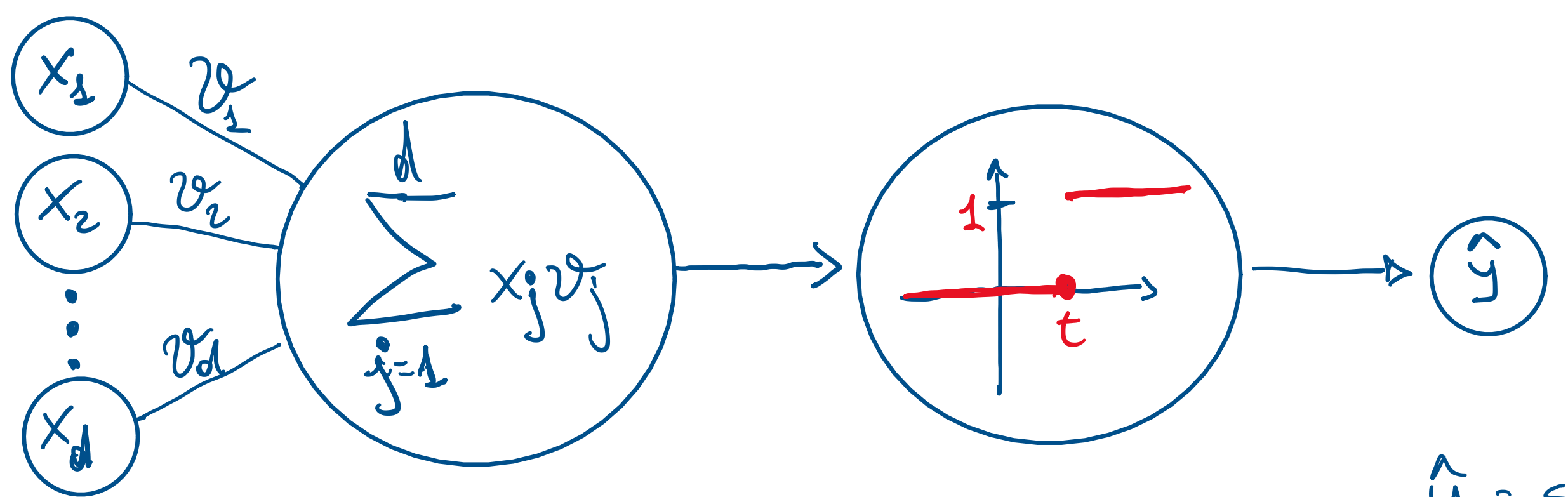
ANY INDUSTRY NEED MACHINE LEARNING EXPERTS
ANY STUDENT/SCIENTIST NEED TO HAVE A GOOD ML BACKGROUND

TODAY
- BETTER HW
- BIGGER DATASETS
- BETTER METHODS

## PERCEPTRON (LIBRO: NEURAL NETWORK - A SYSTEMATIC INTRODUCTION, CAP 3)

PROPOSED BY ROSENBLATT AS MACHINE FOR BINARY CLASSIFICATION ⇒ $x \in \mathbb{R}^d, y \in \{0,1\}$

MAIN IDEA
- ONE WEIGHT $w_j$ ∀ INPUT COMPONENT $x_j$.
- MULTIPLY WEIGHTS WITH RESPECT TO THE INPUTS AND SUM ALL OF THEM (LINEAR COMBINATION)
- IF RESULT IS LARGER THAN A THRESHOLD $t$, THAN RETURN 1 OTHERWISE RETURN 0.


$x_1$
$x_2$
$x_d$
$w_1$
$w_2$
$w_d$
$\sum_{j=1}^{d} x_j w_j$
1
t
$\hat{y}$

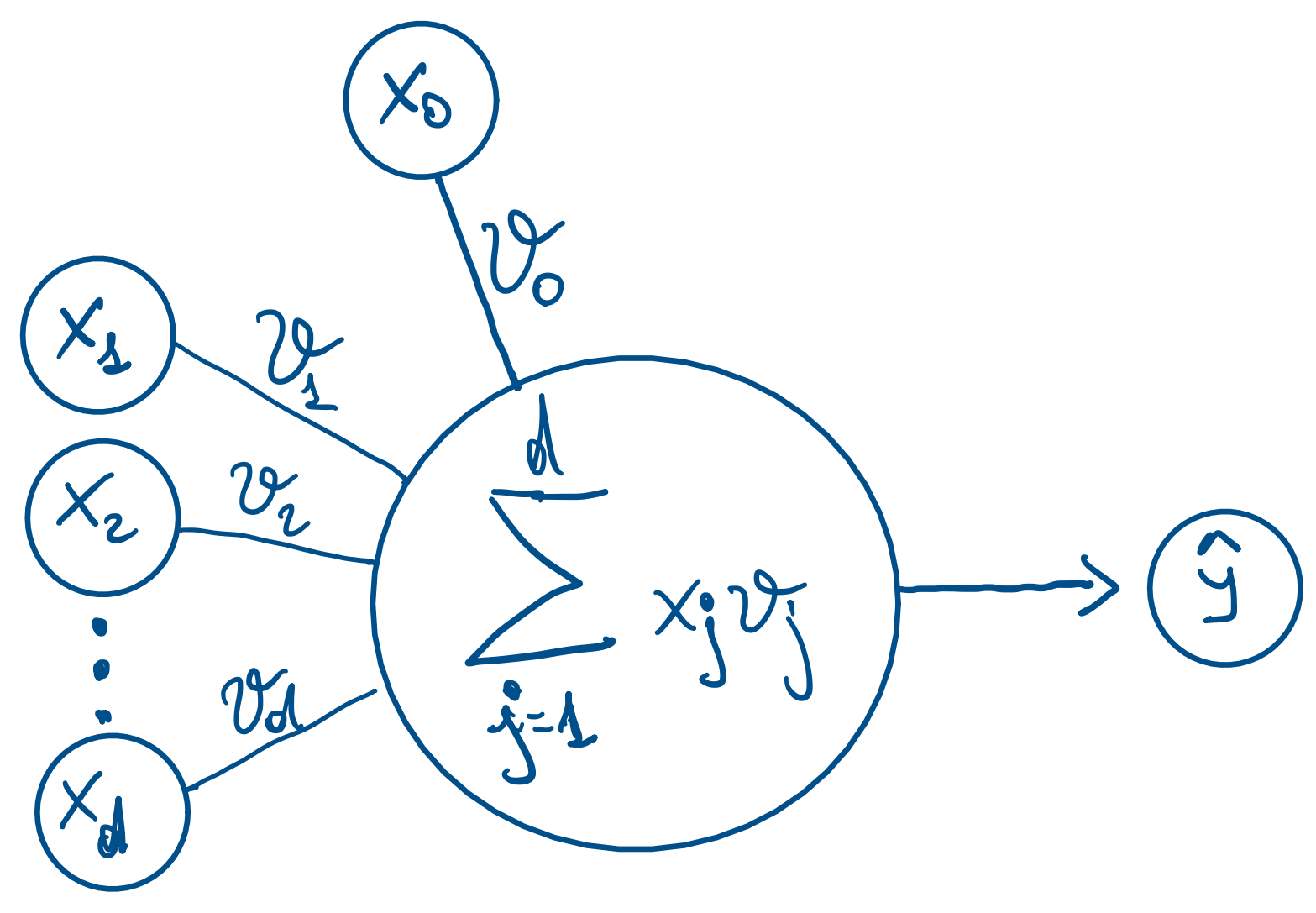
$\hat{y}$ = STIMA

$y$ = ETICHETTA DEL TRAINING

$$\hat{y} = \begin{cases} 1 & \sum_{j=1}^{d} x_j w_j > t \\ 0 & \text{OTHERWISE} \end{cases}$$

↑ PARAMETRIZED SIGN FUNCTION

HOW TO SET THRESHOLD?
- ADD BIAS $x_0 = +1$ AND $w_0$ IN ORDER TO LEARN ALSO THE THRESHOLD


$x_0$
$w_0$
$x_1$
$x_2$
$x_d$
$w_1$
$w_2$
$w_d$
$\sum_{j=1}^{d} x_j w_j$
$\hat{y}$

WE HAVE A MODEL. HOW TO TRAIN IT? HOW TO ADJUST PARAMETERS?

## PERCEPTRON TRAINING (LIBRO: NEURAL NETWORK - A SYSTEMATIC INTRODUCTION, CAP 4)

ROSENBLATT MAIN INNOVATION WAS ON THE TRAINING PROCEDURE FOR PERCEPTRON

### ALGORITHM FOR LEARNING PERCEPTRON

1) INITIALIZE WEIGHTS RANDOMLY WITH SMALL VALUES $\in [0,1]$ AND SET RANDOM THRESHOLD $\in \mathbb{R}$

$$w_j \leftarrow \text{RANDOM}([0,1])$$
$$t \leftarrow \text{RANDOM}(\mathbb{R})$$

2) TAKE SAMPLES $x^{(i)}$ AND PREDICT $\hat{y}^{(i)}$ BY USING PERCEPTRON MODEL

3) FOR ERRONEOUS PREDICTIONS UPDATE WEIGHTS AS FOLLOWS:
- IF THE PREDICTION $\hat{y}^{(i)} = 0$ AND $y^{(i)} = 1$ THEN "INCREASE WEIGHTS"
- IF THE PREDICTION $\hat{y}^{(i)} = 1$ AND $y^{(i)} = 0$ THEN "DECREASE WEIGHTS"

4) REPEAT FROM 2 UNTIL "NO ERRORS ARE MADE"

HOW TO WRITE THIS WITH MATH?

CURRENT VALUE AT TIME T

$$w_j \leftarrow w_j + \underbrace{\left(y^{(i)} - \hat{y}^{(i)}\right)} x_j^{(i)}$$

↑ new $w_j$ at time t+1

← OR WE CAN SUBSTITUTE WITH A SMALL $\varepsilon \in [0,1]$

← WE CAN ASSUME NORMALIZATION IN [0,1]

THESE CAN ASSUME VALUES

| $y^{(i)}$ | $\hat{y}^{(i)}$ |
|---|---|
| 0 | 0 |
| 0 | 1 |
| 1 | 0 |
| 1 | 1 |

IS THE ABOVE PROCEDURE CONVERGING?
WHO IS T?
WHO IS H?
WHICH IS THE MODEL?
WHO IS P?
HOW TO EVALUATE ERRORS DURING TRAINING?

CASE OF PERCEPTRON ↓

COST FUNCTION (LOSS FUNCTION) = COUNT # OF ERRORS $\hat{y}^{(i)} \neq y^{(i)}$ WITH RESPECT TO THE TRAINING SET.

↑ WE WISH TO MINIMIZE

## ADELINE ALGORITHM VARIATION

WEIGHTS CAN BE ADJUSTED WITH A WEIGHTED SUM OF THE INPUT $\hat{y} = \sum_j w_j x_j$